

PAPER_782: NGC 1672 — UQFF Barred Spiral Active Star Formation

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Framework: UQFF (Universal Quantum Field Superconductive Framework)

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CP4 Class: #366 — NGC1672BarredSpiralUQFFCalculator

Abstract

NGC 1672 is a large barred spiral galaxy (SBb) ~60 million light-years away ($z \approx 0.004$) in the constellation Dorado, noted for its remarkably well-defined bar structure and prominent spiral arms containing numerous HII regions and star clusters. It was imaged by Hubble ACS in 2005 and again by JWST in 2023 with extraordinary resolution. NGC 1672 hosts an active nucleus and an above-average star-formation rate for its mass class ($\sim 3 M_{\odot}/\text{yr}$), with its bar efficiently funneling gas toward the central starburst ring. Under UQFF, the elevated bar-driven SFR increases M_{sf} and the outflow velocity to $v = 2 \times 10^5$ m/s, doubling the standard result to $g_{\text{NGC1672}} \approx 2.107 \times 10^{-3} \text{ m/s}^2$.

1. Introduction

NGC 1672's bar is classified as one of the strongest (type SB rather than SAB), indicating efficient gas inflow to the central region. JWST NIRCам and MIRI imaging in 2023 revealed hundreds of star clusters in the spiral arms and the details of the central ring starburst. The combined bar-driven gas flow plus centrally concentrated starburst ring make NGC 1672 a higher-velocity system than symmetric spirals like M74. UQFF captures the bar-driven enhancement through $v = 2 \times 10^5$ m/s (double the symmetric spiral value) and $M_{\text{sf}} = 0.06$, yielding $g_{\text{NGC1672}} = 2.107 \times 10^{-3} \text{ m/s}^2$ — twice the standard result.

2. Master UQFF Gravity Equation

$$g_{\text{NGC1672}}(r, t) = (G \times M) / r^2 \times (1 + H(z) \times t) \times (1 + M_{\text{sf}}) \times (1 + f_{\text{TRZ}}) + a_{\text{EM}}$$

2.1 Parameters

Parameter	Symbol	Value	Source
Galaxy mass	M	$10^{11} M_{\odot} = 1.989 \times 10^{41} \text{ kg}$	HST/JWST
Disk radius	r	$3 \times 10^{20} \text{ m}$ (~32 kly)	NED
SFR (bar-driven)	—	$3 M_{\odot}/\text{yr}$	JWST 2023
Age	t	$5 \times 10^9 \text{ yr} = 1.578 \times 10^{17} \text{ s}$	Hubble time
M_{sf}	—	0.06	UQFF bar-enhanced
Redshift	z	0.004	Spectroscopic
v_{EM}	v	$2 \times 10^5 \text{ m/s}$	Bar-driven outflow
B_{EM}	B	10^{-5} T	Galactic field
f_{TRZ}	—	0.05	UQFF bar

3. Long-Form Derivation

Step 1: Base Gravitational Term

$$g_{\text{grav}} = 6.6743\text{e-}11 \times 1.989\text{e}41 / (3\text{e}20)^2 = 1.476\text{e-}10 \text{ m/s}^2$$

Step 2: Cosmic Expansion Factor

$$H(z) = 2.285\text{e-}18 \text{ s}^{-1}; H(z) \times t = 0.361; \text{ factor} = 1.361$$

Step 3: SFR Mass Fraction (Bar-Enhanced)

$$M_{\text{sf}} = 0.06; 1 + M_{\text{sf}} = 1.06$$

Step 4: Time-Reversal Correction

$$f_{\text{TRZ}} = 0.05; 1 + f_{\text{TRZ}} = 1.05$$

Step 5: Gravitational Total

$$g_{\text{grav_total}} = 1.476\text{e-}10 \times 1.361 \times 1.06 \times 1.05 = 2.237\text{e-}10 \text{ m/s}^2$$

Step 6: Aether EM Correction (Bar-Enhanced v)

$$a_{\text{EM}} = (1.602\text{e-}19 \times 2\text{e}5 \times 1\text{e-}5 / 1.673\text{e-}27) \times 11 \times 1\text{e-}12 = 2.107\text{e-}3 \text{ m/s}^2$$

Step 7: Final Solution

$$g_{\text{NGC1672}} = 2.237\text{e-}10 + 2.107\text{e-}3 \approx 2.107\text{e-}3 \text{ m/s}^2$$

4. Physical Interpretation

NGC 1672's strong SB bar drives gas flow at double the symmetric spiral velocity. JWST 2023 imagery confirms the central starburst ring fed by this bar, validating $v = 2 \times 10^5 \text{ m/s}$ as the UQFF bar-flow parameter. The $2\times$ enhancement relative to standard spirals ($g = 2.107 \times 10^{-3} \text{ vs. } 1.053 \times 10^{-3} \text{ m/s}^2$) directly reflects the bar-driven kinematic intensification captured by UQFF's linear v -dependence.

5. Conclusions

UQFF applied to NGC 1672 yields $g \approx 2.107 \times 10^{-3} \text{ m/s}^2$, exactly double the standard SBbc result. Strong bar-driven gas inflow and the central starburst ring, confirmed by JWST 2023, validate $v = 2 \times 10^5 \text{ m/s}$ as UQFF's bar-drive velocity parameter.

PAPER_782, CP4 class #366. v5.42.

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.